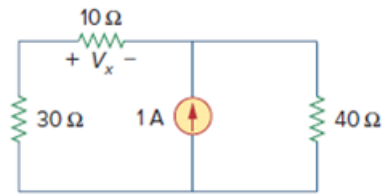


Problem

Use linearity and the assumption that $V_o = 1$ V to find the actual value of V_o in Fig. 4.75.

Figure. 4.75:

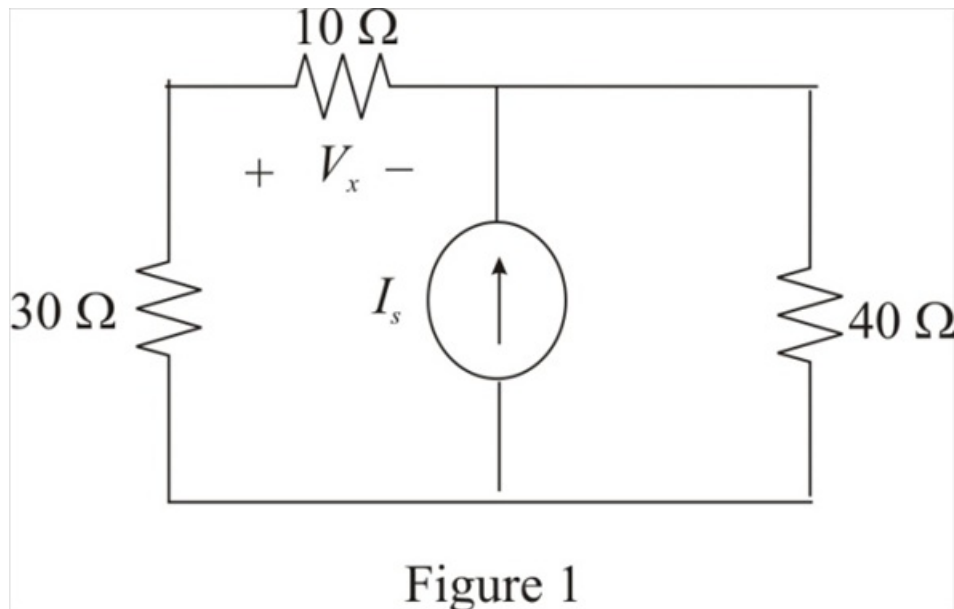


Step-by-step solution

Step 1 of 3

Refer to Figure 4.75 in the text book.

Consider the circuit diagram shown in Figure 1.



Step 2 of 3

Assume the value of the voltage V_o to be V_x .

Calculate the value of voltage V_x using current division rule.

$$\begin{aligned}V_x &= -I_s \left(\frac{40}{40+10+30} \right) (10) \\&= -I_s \left(\frac{40}{80} \right) (10) \\&= -I_s \left(\frac{1}{2} \right) (10) \\&= -5I_s\end{aligned}$$

Substitute 1 V for V_x in the equation.

$$\begin{aligned}1 &= -5I_s \\I_s &= -\frac{1}{5} \\I_s &= -0.2 \text{ A}\end{aligned}$$

Therefore, the value the voltage V_x is 1 V for the source current I_s of -0.2 A .

Step 3 of 3

The actual value of the source current is -5 times the current but not -0.2 A . Due to linearity property, the value of the voltage V_x also increases by -5 times.

Calculate the value of the voltage V_x for $I_s = 1 \text{ A}$.

$$\begin{aligned}V_x &= (1)(-5) \\&= -5 \text{ V}\end{aligned}$$

Therefore, the value of the voltage V_x is -5 V.